Lab 4: Deploying Your Model to Production · Tonight in 20 Steps

OPIM 5512 · Module 3 · Solo · Colab · Dr. Wanik's 453 Hartford cars, then your own Google Cloud bucket

> **The one-sentence version: fit a really good model, deploy it to Google Cloud, and watch it price every new car.** You beat the baseline, tune three model families, check your champion with your own plots, ship it to your bucket, and see the prices (and the SHAP "why") production saves for brand-new cars.

Before we start

- [ ] `predict-shap` **is deploying** (notebook Part 0): Sync fork → **Update branch** (A), or Actions → **Deploy predict-shap** → **Run workflow** (B), or the Cloud Shell block (C)
- [ ] **2-step verification is on** for your Google Cloud Gmail (Google Cloud may ask; click **Enable MFA**)
- [ ] You know your **project ID** and **bucket name** (usually the same)
- [ ] The notebook is open: HuskyCT → In-Class Labs → **Lab 4: Deploying Your Model to Production**

> ⚠️ **The 3 things that trip everyone up (read this first)**
>
> 1. **Save your work:** the notebook opens from GitHub, so first do **File → Save a copy in Drive**.
> 2. **The Google pop-up in 6b: pick the Gmail you used for Google Cloud.** Wrong account = "cannot see the bucket".
> 3. **Part 7 says exactly 500 cars?** You have more than 500 in your bucket. Production prices 500 per run; the rest come at the next :25. Nothing is broken.

The 20 steps, in order

Part 0–1 · Start up and load (steps 1–4, 15 min)

1. Open the notebook from HuskyCT → **File → Save a copy in Drive** (work in the copy).
2. **Part 0:** find your case (A, B or C) and get `predict-shap` deploying. It takes about 3 minutes, so start it now.
3. **1a:** run the install cell (scikit-learn 1.6.1, shap 0.48.0). Asked to restart? Restart, then **Runtime → Run all** later.
4. **1b:** leave `USE_MY_OWN_DATA = False`, type your **PROJECT_ID** and **BUCKET_NAME** → run → **453 cars, 9 columns**.

Part 2 · Clean (steps 5–6, 10 min)

5. **2a:** see the mess (`Ford` vs `ford` , `chevy` vs `chevrolet`), then run `clean_cars` : one name per brand.
6. **2b:** run the flags (odd price, placeholder miles, very old) → **6 flagged, 447 ready**. Read why each one was flagged.

Part 3 · Baseline (steps 7–8, 10 min)

7. **3a–3b:** split (**357 training / 90 test**), then **peek inside the recipe**: 4 inputs become **98 columns**.
8. **3c–3d:** the numbers to beat: guess the median, then the plain tree (`train_cv_mae` **± std**, and `test_mae`).

Part 4 · Tune (steps 9–11, 25 min)

9. Run the grid search (1–2 minutes): a decision tree, a random forest and KNN, each scored with 5-fold CV.
10. **Class challenge:** add values to any grid, re-run, and try for the lowest `train_cv_mae` . Post your best!
11. **Your champion** is picked for you (lowest `train_cv_mae`). Want another? Uncomment the second line.

Your turn · You write the code (steps 12–13, 15 min)

12. Run the setup cell, then code **Task 1** (actual vs predicted scatter) and **Task 2** (error histogram), champion vs baseline.
13. Code **Task 3** (summary stats table). Talk it over: where does your champion beat the baseline?

Part 5 · Explain (steps 14–15, 20 min)

14. Run the two helpers, then SHAP on your test cars: the **bar chart** and the **beeswarm** (the global picture).
15. **One car** (waterfall), then **5b: price a car you know**. Never-seen make? It says so (the "infrequent" bucket).

Part 6 · Ship it (steps 16–18, 20 min)

16. **6a:** save `model.joblib` and `model_card.json` (the label production reads).
17. **6b:** sign in (pick your **Google Cloud Gmail**) → 3 checks: account, bucket, `predict-shap` ACTIVE. **6c:** upload.
18. **6d:** run `predict-shap` now → dots → **"Done: predict-shap wrote your predictions"** + your two file paths.

Part 7 · Your model at work (step 19, 5 min)

19. **7a:** run it (plumbing). **7b:** the newest cars your model never saw, with its prices **and** SHAP values. Click the link and open `preds_master.csv` and `shap_master.csv` in Google Cloud.

Part 8 · Wrap up (step 20, 5 min)

20. Answer the **five short questions** → **File** → **Download** → **.ipynb** → upload to **Submit Lab 4** (due **Thu Oct 22, 11:59 PM**, Hartford and Stamford).

Your two production files (they grow every hour at :25)

File	What is in it
`gs://YOUR-BUCKET/structured/preds_lab4/preds_master.csv`	every car, its asking price, your model's price, the error, the model version
`gs://YOUR-BUCKET/structured/preds_lab4/shap_master.csv`	the base value and the 4 SHAP values behind each price

If you get stuck

You see	Do this
"Type your champion..." or a missing variable	Run every cell above it first (Runtime → Run before)
6b: "cannot see the bucket"	Runtime → Disconnect and delete runtime, run again, pick your Google Cloud Gmail
6b: "predict-shap is not ACTIVE"	Part 0: look for a red X in Actions (A/B), or re-run the Cloud Shell block (C)
6d: "Nothing new after 4 minutes"	Run the optional log cell under 6d and paste the log into Claude
Part 7: "DEMO MODE"	Run 6d first, then re-run 7a

Leave `predict-shap` running. Every hour it prices and explains your newest cars; that is your midterm data.